Risk Analysis in Psychological Emergencies through Latent Variable Models

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Abstract

This work describes the application of advanced statistical models in the analysis of data from the research project "Analysis of emergencies admitted to the Psychology Service of the 911 Emergency System of Salta," directed by Lic. Carolina Rojo. A questionnaire called the General Protocol for Psychological Emergencies was designed to evaluate various types of consultations, particularly psychological emergencies. This study implements Rasch models and Latent Classes as fundamental statistical tools to explore the dimensionality of the Protocol items, facilitating psychology professionals in evaluating the obtained results. The analyzed data corresponds to the population that contacts the Psychology Service of the 911 Emergency System of Salta province.

The methodology employed consisted of applying the Rasch model and the mixed Rasch model, with subsequent removal of poorly fitting items and evaluation of the Protocol's unidimensionality. The initial Rasch analysis of twelve items identified six items with poor fit. The elimination of these items produced a six-item instrument that did not present unidimensionality. The mixed Rasch analysis of these six items revealed two distinct unidimensional structures: one associated with individual violence toward the environment and another related to subject violence toward themselves. These structures align with the results of an exploratory factor analysis previously performed on the same evaluation instrument.

Keywords: Rasch model, mixed Rasch model, latent class analysis, psychological emergencies, psychometric evaluation.

1 Introduction

Latent variable models constitute a fundamental tool in the analysis of psychological and social data, allowing the measurement of constructs that are not directly observable. In

the context of psychological emergencies, these models offer the possibility of identifying underlying patterns in consultations and establishing risk profiles that facilitate professional intervention.

This study is framed within the research project directed by Lic. Carolina Rojo, focused on the analysis of emergencies admitted to the Psychology Service of the 911 Emergency System of Salta. The relevance of this work lies in the need to develop psychometrically sound evaluation instruments that allow efficient categorization of psychological emergencies, thus optimizing the emergency service response.

1.1 Objectives

The main objectives of this study are:

1. Evaluate the psychometric properties of the General Protocol for Psychological Emergencies through latent variable models.
2. Explore the instrument's dimensionality using the Rasch model and its extensions.
3. Identify latent structures in the data that allow better understanding of psychological emergency types.
4. Provide empirical evidence for optimization of the evaluation protocol.

2 Theoretical Framework

2.1 Latent Variables in Psychology

In the fields of psychology and social sciences, latent variable models emerged from the need to measure unobservable constructs or dimensions, such as human intelligence, political inclinations, and specific abilities for performing certain tasks **Ip2003**. Although these types of variables cannot be measured directly, there are frequently related observable variables, called manifest variables, that can be measured and used to infer about latent variables.

Definition 2.1 (Latent Variable)**.** A latent variable is a hypothetical construct that cannot be directly observed, but whose existence is inferred from observable or manifest variables that are assumed to be causally related to it.

2.2 The Rasch Model

The Rasch model, formally belonging to latent trait models, is used to analyze evaluation data with the objective of measuring variables such as attitudes, abilities, and personality traits from responses to questions in an evaluation **Prieto2003**.

2.2.1 Mathematical Foundations

The model is based on the following assumptions:

i. The attribute to be measured can be represented in a single dimension where persons and items would be jointly situated (unidimensionality).

ii. The person's level in the attribute and the item difficulty determine the probability that the response is correct.

The probability of correctly answering a question is modeled as a logistic function of the difference between the person parameter and the item parameter:

$$P_{is} = \frac{e^{(\theta_s - \beta_i)}}{1 + e^{(\theta_s - \beta_i)}} \tag{2.1}$$

where:

- P_{is}: Probability that person s answers item i correctly
- θ_s: Ability level of person s
- β_i: Difficulty level of item i

Solving equation (2.1), we obtain:

$$\ln\left(\frac{P_{is}}{1 - P_{is}}\right) = \theta_s - \beta_i \tag{2.2}$$

2.2.2 Scaling and Calibration

When applying the Rasch model, items are placed on a scale (difficulty estimation), a process known as item calibration. The scalar values of persons and items can be expressed on the logit scale, according to equation (2.1).

Observation 2.1. The location of point 0 is situated at the mean difficulty of the items. In this case, interpretation of person parameters is very simple: values of θ_s greater than 0 indicate that persons have a probability greater than 0.50 of success on items of average difficulty.

2.2.3 Parameter Estimation

Parameter estimation in the Rasch model can be performed through different methods:

a) **Conditional Estimation**: When item parameters are known and person parameters are to be estimated.

b) **Joint Estimation**: When both item and person parameters are unknown.

The most commonly used method is maximum likelihood, which consists of determining the parameters that make the observed responses most probable.

2.2.4 Model Fit

To evaluate data fit to the model, various indices are used:

- **Q-index**: Fit statistic that varies between 0 and 1. Values close to 0 indicate perfect discrimination, while values close to 0.50 indicate random responses. Values between 0 and 0.30 are expected.
- Z_q: Transformation of the Q-index with approximately normal distribution. Values between -2 and 2 are considered acceptable.
- **RMSEA (Root Mean Square Error of Approximation)**: Goodness of fit index to evaluate unidimensionality. Values ≤ 0.10 indicate good fit, ≤ 0.05 superior fit, and ≤ 0.01 outstanding fit.

2.3 Latent Class Analysis

The latent class model is a class of model based on unsupervised learning, whose objective is the self-organization of the record collection into a predetermined number of homogeneous groups **Magidson2004**.

Definition 2.2 (Latent Class)**.** A latent class is characterized by a conditional probability pattern that indicates the probability that observed variables take certain values, given that the individual belongs to that class.

The criteria for solving latent class analysis are based on achieving classes within which there is no association between observed variables (conditional independence).

2.4 Mixed Rasch Model

The mixed Rasch model extends the standard Rasch model by formally integrating latent class analysis **Alexeev2010**. The fundamental idea is that if the Rasch model is not fulfilled for a dataset, the reason may reside in the sample consisting of more than one group of subjects. Within each group, the Rasch model is fulfilled, but classes differ in rank order and in item difficulty differences.

Theorem 2.1 (Mixed Rasch Model)**.** If a heterogeneous population can be divided into K homogeneous subpopulations, then in each subpopulation k the Rasch model is fulfilled with specific parameters $\theta_{s|k}$ and $\beta_{i|k}$.

2.5 Classification of Latent Variable Models

Latent variable models can be classified according to the nature of observed and latent variables:

Table 1: Classification of latent variable models

Latent Variable	**Observed Variable**	
	Discrete	Continuous
Discrete	Latent class models	Finite mixture models
Continuous	Latent trait models	Factor analysis models

3 Methodology

3.1 Study Design

A cross-sectional observational study with psychometric analysis of the General Protocol for Psychological Emergencies applied in the Psychology Service of the 911 Emergency System of Salta was conducted.

3.2 Sample

The sample consisted of 310 protocols completed by service personnel during psychological emergency calls. The data corresponds to the actual population that contacts the emergency service.

3.3 Instrument

The General Protocol for Psychological Emergencies consists of 12 dichotomous items that evaluate different aspects of the subjective context of the emergency:

1. Current suicide attempt
2. Previous suicide attempts
3. Suicidal ideation
4. Self-harm
5. Addiction
6. Abandonment
7. Separation
8. Difficulty setting limits
9. Psychiatric patient

10. Sexual problems
11. Distress
12. Episodes of violence

3.4 Analysis Procedure

Data analysis was performed following these stages:

3.4.1 Stage 1: Latent Class Analysis

Latent class analysis was applied to identify homogeneous subpopulations. Models with different numbers of classes were evaluated using the Bayesian information criterion (BIC).

3.4.2 Stage 2: Rasch Model Application

The Rasch model was applied to the complete sample, evaluating:

- Item fit through Q-index and Z_q
- Person fit
- Unidimensionality through RMSEA

3.4.3 Stage 3: Mixed Rasch Model

Given the poor fit of the standard Rasch model, we proceeded to apply the mixed Rasch model, evaluating different numbers of classes.

3.4.4 Stage 4: Model Optimization

An iterative process was performed:

1. Elimination of poorly fitting items
2. Model re-evaluation
3. Application of mixed Rasch model with selected items

3.5 Software Used

Analyses were performed using:

- Winmira for Rasch models
- Microsoft Excel for descriptive analyses
- XLSTAT for exploratory factor analysis with tetrachoric correlations

4 Results

4.1 Latent Class Analysis

The initial latent class analysis indicated that the optimal model was three classes according to the BIC criterion:

Table 2: Information criteria for different numbers of classes

Number of Classes	CAIC	BIC
1	3003	2991
2	2911	2886
3	2923	2885
4	2959	2908

The three identified classes presented the following characteristics:

4.1.1 Class 1 (74% of the sample)

This class was characterized by low affirmative response probabilities in all items. The most relevant were:

- Distress (0.42)
- Episodes of violence (0.35)
- Psychiatric patient (0.28)

4.1.2 Class 2 (20% of the sample)

Presented high probabilities in:

- Current suicide attempt (0.50)
- Suicidal ideation (0.55)
- Distress (0.35)

4.1.3 Class 3 (6% of the sample)

Showed high probabilities in most items:

- Distress (0.94)
- Episodes of violence (0.81)

- Suicidal ideation (0.75)
- Psychiatric patient (0.63)

4.2 Rasch Model Application

The initial application of the Rasch model to the 12 items showed:

Table 3: Item fit to initial Rasch model

Item	Score	Q-index
Current suicide attempt	0.16	0.4016*
Previous suicide attempts	0.07	0.0720
Suicidal ideation	0.18	0.2210
Self-harm	0.05	0.2060
Addiction	0.20	0.2465
Abandonment	0.04	0.5109*
Separation	0.06	0.2340
Difficulty setting limits	0.15	0.4383*
Psychiatric patient	0.25	0.4248*
Sexual problems	0.04	0.5852*
Distress	0.44	0.2388
Episodes of violence	0.35	0.2654

*Items with Q-index > 0.30 (poor fit)

RMSEA = 0.383 indicated poor fit of the unidimensional model.

4.3 Mixed Rasch Model

Application of the mixed Rasch model with 2 classes showed:

- Class 1: RMSEA = 0.347
- Class 2: RMSEA = 0.455

Both values indicated inadequate fit, leading to the optimization strategy.

4.4 Model Optimization

After eliminating items with Q-index > 0.30 (Abandonment, Difficulty setting limits, Sexual problems), 6 items considered conceptually relevant were maintained. The mixed Rasch model with 3 classes showed:

Table 4: Fit of optimized mixed Rasch model

Class	n (%)	RMSEA	Fit
1	254 (82%)	0.106	Acceptable
2	43 (14%)	0.174	Mediocre
3	13 (4%)	2.268	Rejectable

4.5 Dimension Interpretation

Analysis of discriminative items in each class allowed identification of two main dimensions:

4.5.1 Dimension 1: Violence toward the environment

Characterized by items:

- Distress
- Episodes of violence

This dimension predominated in Class 1 (82% of the sample).

4.5.2 Dimension 2: Violence toward oneself

Characterized by items:

- Suicidal ideation
- Current suicide attempt (Class 2)
- Previous suicide attempts and Addiction (Class 3)

4.6 Validation with Factor Analysis

The previous exploratory factor analysis identified three factors that aligned with the found dimensions:

1. **Factor 1**: Subject's outcome behaviors toward themselves (suicidal ideation, self-harm, addiction)
2. **Factor 2**: Experienced emotional exposure (abandonment, separation, sexual problems)
3. **Factor 3**: Outcome behaviors toward the environment (episodes of violence, difficulty setting limits)

The convergence between mixed Rasch model results and factor analysis provides construct validity evidence.

5 Discussion

5.1 Theoretical Implications

The obtained results have important implications for understanding psychological emergencies:

5.1.1 Construct Multidimensionality

The rejection of the unidimensional Rasch model and the need to resort to the mixed model indicate that psychological emergencies do not constitute a unidimensional construct. This heterogeneity suggests the existence of different types of emergencies that require differentiated approaches.

5.1.2 Emergency Typology

The identification of two main dimensions (violence toward the environment and toward oneself) provides an empirically grounded typology that can guide professional intervention.

5.2 Practical Implications

5.2.1 Protocol Optimization

The reduction from 12 to 6 items without significant loss of information suggests the possibility of developing a more efficient instrument. The eliminated items (abandonment, separation, sexual problems, difficulty setting limits) showed low discriminative power in the emergency context.

5.2.2 Intervention Protocols

The identification of differentiated profiles suggests the need to develop specific intervention protocols:

- For "violence toward environment" type emergencies: focus on containment and aggression management
- For "violence toward oneself" type emergencies: suicide risk evaluation and activation of specific protocols

5.3 Study Limitations

1. **Sample size**: Although adequate for main analyses, the size of Class 3 (n=13) limits conclusions about this subgroup.

2. **Dichotomous nature of items**: Dichotomization may result in loss of information about symptom severity.

3. **Specific context**: Results are limited to the context of the 911 Emergency System of Salta and may not be generalizable to other contexts.

4. **Absence of external validation**: External criteria were not available to validate the obtained classifications.

5.4 Future Directions

5.4.1 Instrument Development

The development of polytomous versions of the protocol that capture symptom severity, not just presence/absence, is recommended.

5.4.2 Validation Studies

Studies are needed that:

- Validate the identified dimensions with external criteria
- Evaluate temporal stability of classifications
- Examine predictive validity regarding clinical outcomes

5.4.3 Theoretical Framework Expansion

Integration of these findings with contemporary theories about psychological emergencies and crisis intervention models could enrich understanding of the phenomenon.

6 Conclusions

This study demonstrates the utility of latent variable models for psychological emergency analysis. The main findings include:

1. Psychological emergencies attended by the 911 Emergency System of Salta do not constitute a unidimensional phenomenon, but present at least two main dimensions: violence toward the environment and violence toward oneself.

2. The General Protocol for Psychological Emergencies can be optimized, reducing the number of items from 12 to 6 without significant loss of diagnostic information.

3. There is notable convergence between mixed Rasch model results and exploratory factor analysis, providing construct validity evidence.

4. The majority of emergencies (82%) correspond to a lower severity profile characterized mainly by distress and episodes of violence toward the environment.

5. A smaller but significant percentage (14%) presents a high-risk profile characterized by current suicidal ideation and attempts.

These findings have important implications for professional practice in psychological emergency services, suggesting the need for differentiated evaluation and intervention protocols according to the type of emergency identified. The application of advanced psychometric models such as the mixed Rasch model allows a deeper and more nuanced understanding of complex phenomena such as psychological emergencies, facilitating the development of more effective and efficient interventions.

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